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PRISM: A Probabilistic Regime-Integrated Scalping Model for Derivatives Arbitrage

A Unified Framework for Gamma Scalping, Markov Regime
Dynamics, Futures Basis Arbitrage, and Stochastic Control
in Equity Derivatives Markets

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Abstract

This paper introduces PRISM, the Probabilistic Regime-Integrated Scalping Model, the proprietary derivatives arbitrage engine developed by the VRC derivatives research team. PRISM is to equity derivatives what AEGIS is to equity valuation: it operationalizes VRC’s core philosophical conviction—that edge in financial markets is not a static number but a continuously updated probability distribution—in the specific context of volatility arbitrage. PRISM formalizes the observation that gamma scalping is not merely a bet on realized versus implied volatility, but a *regime-conditional arbitrage* whose edge, execution, and risk profile depend fundamentally on the latent volatility state of the market. At VRC, we do not trade gamma scalping the way textbooks describe it; we trade it the way PRISM prescribes it. The framework rests on five pillars developed and calibrated by VRC: (i) a Markov regime-switching geometric Brownian motion (MRSGBM) for the underlying, calibrated to the VRC derivatives coverage universe; (ii) a coupled PDE system for option pricing across regimes, solved numerically on VRC’s finite-difference grid; (iii) a three-asset no-arbitrage theorem connecting options, futures, and spot, which characterizes the structural edge available to the VRC derivatives desk; (iv) a Hamilton-Jacobi-Bellman formulation of the optimal hedging bandwidth problem incorporating VRC’s observed transaction cost structure; and (v) a multi-signal particle filter engine for real-time regime detection, augmented with the options surface likelihood that is VRC’s key innovation. The central result establishes that the discounted gamma scalping P&L decomposes into a sum of regime-weighted variance differences, with weights determined by the regime-conditional dollar gamma profile of the option. Embedding this into a three-asset system including a mean-reverting futures basis process, VRC proves that futures-based hedging dominates spot hedging in transaction-cost-adjusted terms and provides a complementary mean-reversion income stream through basis drift. VRC solves the associated HJB equation in closed form for the optimal rebalancing threshold, derives the Fokker-Planck equation governing the distribution of cumulative P&L, and obtains Kelly-optimal position sizes with an entropy-based uncertainty penalty that automatically scales VRC’s allocations down during regime ambiguity. Numerical simulations on the VRC-calibrated three-regime model demonstrate annualized Sharpe ratios of 1.6–2.4 after transaction costs, with the VRC particle filter achieving regime identification accuracy exceeding 89% and detection latency under two trading days. A

complete VRC trading protocol with entry conditions, hedge execution rules, roll management, and risk controls constitutes the operational backbone of the VRC derivatives book.

Keywords: gamma scalping, Markov regime-switching, stochastic volatility, delta hedging, realized vs. implied variance, futures basis arbitrage, HJB equation, Fokker-Planck, particle filter, Kelly criterion, options market microstructure, volatility surface.

JEL Classification: G13, G12, G11, C61, C65, C11.

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1. Introduction

The relationship between the volatility implied by option prices and the volatility subsequently realized by the underlying asset is one of the most studied phenomena in quantitative finance, and one of the most persistently profitable to trade. The empirical regularity that implied volatility systematically exceeds subsequently realized volatility—the so-called *variance risk premium* (VRP)—has been documented across equity indices, individual equities, foreign exchange, and fixed income markets [Carr and Wu, 2009, Bollerslev et al., 2009]. The VRC derivatives team has observed this premium firsthand across the SPX, NDX, and single-stock option markets that constitute our primary coverage universe. The gamma scalping strategy—long options, continuously delta-hedged—is the canonical mechanism for harvesting this premium: its P&L is proportional to $\sigma_t^2 - \hat{\sigma}^2$ per unit of dollar gamma, so the strategy earns money precisely when realized volatility exceeds implied, a condition VRC has found to hold more often than not when entered under the right regime conditions.

Yet the practical execution of gamma scalping has remained more art than science. The classical Black-Scholes P&L identity, which tells the VRC desk *that* there is an edge, provides no guidance on *how large* the position should be, *how frequently* to rebalance given VRC’s transaction cost structure, *which instrument*—spot or futures—to use for the delta hedge, or *how to respond* when the market shifts from a low-volatility to a high-volatility regime mid-trade. These are the questions that PRISM answers, and that VRC now operationalizes daily.

1.1 The Three Gaps This Paper Fills

Gap 1: Regime Structure. Volatility is not a stationary scalar, and VRC’s own data confirms this. The VRC derivatives desk has identified at least three distinct volatility regimes in US equity markets over the 2000–2024 period: a calm regime (realized vol 9–13%), a normal regime (18–28%), and a crisis regime (38–55%). Each regime has distinct entry characteristics, optimal bandwidth prescriptions, and expected P&L profiles. A flat-vol model that averages across regimes will over-trade in low-vol periods (buying expensive rebalances when gamma income is thin) and under-trade in high-vol periods (missing the bulk of the opportunity). PRISM, which VRC uses as the live regime-detection and bandwidth engine, addresses this by modeling the underlying as a Markov regime-switching geometric Brownian motion and deriving the exact regime-conditional decomposition of the gamma scalping P&L.

Gap 2: Futures Integration and Basis Arbitrage. The VRC derivatives desk,

consistent with institutional best practice, uses futures—not spot—to hedge delta on all listed equity derivatives positions. The reasons are well understood at VRC: E-mini futures require no cash outlay, carry bid-ask spreads of 0.1–0.3 bps versus 3–10 bps for spot blocks, and allow intraday margin netting across the VRC portfolio. What is less appreciated, even within VRC, is that the futures basis—the deviation of the traded E-mini price from its theoretical cost-of-carry value—is itself a mean-reverting stochastic process that generates an additional, identifiable P&L stream for the futures-hedging gamma scalper. This *basis arbitrage enhancement*, which VRC now captures systematically, has not been formally characterized in the gamma scalping literature. We prove it, quantify it, and show precisely how the VRC execution desk should position around it.

Gap 3: Optimal Control and Position Sizing. Given regime uncertainty, VRC’s transaction cost structure, and the non-Gaussian distribution of cumulative P&L, what is the optimal strategy for the VRC gamma scalper? We answer this by formulating and solving the Hamilton-Jacobi-Bellman (HJB) equation for the joint state $(S_t, \delta_t, b_t, X_t)$ —where δ_t is the current delta mismatch and b_t is the futures basis—and by deriving the Fokker-Planck equation for the distribution of cumulative P&L. Kelly-optimal position sizing with an entropy penalty on regime uncertainty closes the loop between theory and the VRC execution desk, giving the portfolio manager a single number: the correct position size at any moment, given everything VRC knows about the current regime.

1.2 PRISM Within the VRC Research Canon

PRISM is the derivatives complement to AEGIS [Verma, 2026a], VRC’s proprietary equity valuation platform. Where AEGIS builds a probabilistic intrinsic value engine for global equities and operates as VRC’s equity research backbone, PRISM builds a probabilistic volatility arbitrage engine for equity derivatives and operates as VRC’s derivatives trading backbone. Both systems are expressions of the same VRC investment philosophy articulated in Verma [2026b]: that financial signals are best represented as probability distributions continuously refined by new information, that no model output should ever be treated as a point estimate, and that the VRC analyst’s edge lies not in predicting the future but in maintaining a disciplined probabilistic framework when the market abandons one.

Practically speaking, PRISM and AEGIS interact: when AEGIS flags a company as materially undervalued on the VRC composite score Ω , the VRC investment committee reviews whether buying calls on that name (rather than the underlying) is the better risk-adjusted entry. PRISM provides the volatility pricing assessment—

whether the market’s implied vol for that name is cheap or expensive relative to VRC’s regime-conditional realized vol estimate—that informs that decision. The two systems together constitute VRC’s integrated equity and derivatives research infrastructure.

1.3 Organization of the Paper

Section 2 establishes the theoretical foundations. Section 3 develops the MRSGBM model and option pricing PDE system. Section 4 derives the full regime-switching P&L identity. Section 5 introduces futures dynamics and basis processes. Section 6 proves the three-asset arbitrage theorems. Section 7 solves the optimal bandwidth HJB problem. Section 8 derives the P&L distribution and Kelly sizing. Section 9 presents the particle filter regime detection engine. Section 10 develops the variance risk premium forecasting model. Section 11 addresses volatility surface dynamics and skew. Section 12 discusses microstructure and execution. Section 13 provides numerical results and backtests. Section 14 gives the complete practical trading protocol. Section 15 discusses extensions and future development. Section 16 describes limitations and risk disclosures. Section 17 concludes.

2. Theoretical Foundations

2.1 The Gamma Scalping P&L as a Convexity Trade

The intellectual foundation of PRISM begins with the most basic result in options replication theory, one that every VRC derivatives analyst internalizes on day one: the P&L of a continuously delta-hedged long option position depends not on the direction of the underlying but on the realized quadratic variation of its price process. Direction is for equity analysts. Gamma is for derivatives.

The standard Black-Scholes P&L identity (see Wilmott 2006) states that for a European option with value $V(S, t)$, continuously delta-hedged, the instantaneous P&L satisfies:

$$d\Pi_t = \underbrace{\frac{1}{2} S_t^2 \Gamma_t (\sigma_t^2 - \hat{\sigma}^2)}_{\text{variance P\&L}} dt + \underbrace{r \Pi_t}_{\text{carry}} dt, \quad (1)$$

where σ_t is the instantaneous realized volatility, $\hat{\sigma}$ is the Black-Scholes implied volatility at which the option was purchased, and $\Gamma_t = \partial^2 V / \partial S^2$ is the dollar gamma.

Equation (1) is elegant but misleading in one important respect: it presents the P&L as if volatility were a fixed parameter. In the real markets that VRC trades, it is not.

VRC’s proprietary realized volatility data shows that the S&P 500 moves through distinct volatility regimes separated by transition events—macro shocks, earnings seasons, Fed meetings, geopolitical developments—that cannot be described by any single scalar σ . This is the central inadequacy of the flat-vol model that PRISM corrects.

2.2 VRC’s Bayesian Framework for Volatility Regime Estimation

The second pillar of PRISM—and of VRC’s broader investment philosophy [Verma, 2026b]—is Bayesian inference. VRC does not forecast which volatility regime will prevail tomorrow; VRC maintains a continuously updated probability distribution over all possible regimes and acts proportionally to it. This is the same philosophical foundation that underlies AEGIS’s treatment of intrinsic value as a distribution rather than a point estimate, applied now to the volatility state space.

Formally, let $\pi_{i,t} = \mathbb{P}(X_t = i \mid \mathcal{F}_t^{\text{obs}})$ denote VRC’s posterior probability of regime i given all observable information (price history, options chain, macroeconomic data) up to time t . VRC updates this posterior continuously via Bayes’ theorem:

$$\pi_{i,t+\Delta t} = \frac{L(E_{t+\Delta t} \mid X_{t+\Delta t} = i) \sum_j P_{ji}(\Delta t) \pi_{j,t}}{\sum_k L(E_{t+\Delta t} \mid X_{t+\Delta t} = k) \sum_j P_{jk}(\Delta t) \pi_{j,t}}, \quad (2)$$

where $P_{ji}(\Delta t) = \exp(\Lambda \Delta t)_{ji}$ is the transition probability from regime j to regime i over Δt , and $L(\cdot \mid \cdot)$ is the emission likelihood. This is the discrete-time version of the Zakai equation for the unnormalized filter; its continuous-time limit is the Kushner-Stratonovich equation described in Section 9.

2.3 The Variance Risk Premium: VRC’s Structural Source of Alpha

The variance risk premium (VRP)—implied variance exceeding expected realized variance—is the structural foundation of every option position VRC takes on the long gamma side. VRC’s analysis of the S&P 500 options market from 2000 to 2024 confirms that the VRP has been positive on average in 64% of monthly observations, with the premium concentrated overwhelmingly in the calm and normal regimes. In crisis regimes, the VRP frequently inverts—realized vol spikes above implied—which is precisely why PRISM’s regime-adaptive position sizing reduces or eliminates long gamma exposure during detected crisis states.

The theoretical motivation for a positive VRP was established by Carr and Wu [2009]: under the representative agent model with Epstein-Zin preferences:

$$\hat{\sigma}^2 - \mathbb{E}^{\mathbb{P}}[\hat{r}\hat{v}^2] = -\text{Cov}^{\mathbb{Q}}(\hat{r}\hat{v}^2, M_{t,T}/M_{t,T}^f), \quad (3)$$

where $M_{t,T}$ is the stochastic discount factor and $M_{t,T}^f$ is the risk-free discount factor. When variance is high (market stress), the SDF is also high (investors are desperate for consumption), creating a negative covariance that drives the VRP positive. VRC exploits this premium systematically: PRISM buys options when their implied vol is below VRC's regime-conditional realized vol estimate, and the VRC investment committee requires that this condition be satisfied—formalized as the PRISM arbitrage condition in Section 6—before any long gamma position is initiated on the VRC derivatives book.

2.4 The Options Market as a VRC Regime Signal

A key innovation of PRISM relative to standard regime-switching filters is VRC's use of the *options market itself* as a regime detection input. The VRC derivatives desk observes the full SPX and single-stock options chain in real time; rather than treating this as merely the output of the valuation model, PRISM feeds the options surface *back in* as an informative prior on the current regime. The volatility surface—specifically, the term structure of at-the-money implied vol, the put skew, and the term structure of VIX futures—embeds the market's collective posterior over future volatility regimes, and VRC reads that information using the Breeden-Litzenberger formula [Breeden and Litzenberger, 1978]:

$$p^{\mathbb{Q}}(S_T \leq K) = e^{rT} \frac{\partial C}{\partial K}, \quad (4)$$

which allows recovery of the risk-neutral density from the options chain. The first and second moments of this density under the VRC regime-switching model provide an independent cross-check on the VRC particle filter's regime estimates, and in practice VRC finds that the options surface signals regime transitions approximately 1.8 trading days *earlier* on average than the returns-only filter (Table 4).

3. Markov Regime-Switching Geometric Brownian Motion

3.1 Probability Space and the Markov Chain

Let $(\Omega, \mathcal{G}, \mathbb{P})$ be a complete probability space supporting a standard Brownian motion $W = (W_t)_{t \geq 0}$ and an independent continuous-time Markov chain $X = (X_t)_{t \geq 0}$ with finite state space $\mathcal{S} = \{1, \dots, N\}$ and generator matrix $\Lambda = (\lambda_{ij})$:

$$\lambda_{ij} \geq 0 \text{ for } i \neq j, \quad \lambda_{ii} = - \sum_{j \neq i} \lambda_{ij}. \quad (5)$$

The joint filtration is $\mathcal{F}_t = \mathcal{F}_t^W \vee \mathcal{F}_t^X$. We write $\mu_i = \mu(i)$ and $\sigma_i = \sigma(i)$ for regime-dependent parameters.

Assumption 3.1 (Irreducibility and Ordering). The chain X is irreducible. Regimes are labeled so that $0 < \sigma_1 < \sigma_2 < \dots < \sigma_N$, i.e., regime 1 is the lowest-volatility state and regime N is the highest.

3.2 Spot Price Dynamics

Definition 3.2 (MRSGBM). The spot price S_t satisfies:

$$\frac{dS_t}{S_t} = \mu_{X_t} dt + \sigma_{X_t} dW_t, \quad S_0 > 0. \quad (6)$$

By Itô's lemma, the log-price $L_t = \ln S_t$ satisfies:

$$dL_t = \left(\mu_{X_t} - \frac{1}{2} \sigma_{X_t}^2 \right) dt + \sigma_{X_t} dW_t. \quad (7)$$

The quadratic variation of L over $[0, t]$ is:

$$\langle L \rangle_t = \int_0^t \sigma_{X_s}^2 ds = \sum_{i=1}^N \sigma_i^2 \tau_i(t), \quad (8)$$

where $\tau_i(t) = \int_0^t \mathbf{1}_{\{X_s=i\}} ds$ is the occupation time of regime i .

3.3 Stationary Properties and the Ergodic Theorem

The unique stationary distribution $\boldsymbol{\pi} = (\pi_1, \dots, \pi_N)$ satisfies $\boldsymbol{\pi} \Lambda = \mathbf{0}$, $\sum_i \pi_i = 1$, $\pi_i > 0$. By the ergodic theorem for irreducible continuous-time Markov chains:

$$\frac{\tau_i(T)}{T} \xrightarrow{T \rightarrow \infty} \pi_i \quad \mathbb{P}\text{-a.s.}, \quad (9)$$

so the long-run realized variance converges to the stationary mixture:

$$\sigma_{\text{RV}, \infty}^2 = \sum_{i=1}^N \pi_i \sigma_i^2. \quad (10)$$

3.4 Option Pricing: The Coupled PDE System

Let $V_i(S, t)$ denote the price of a European derivative with payoff $\Phi(S_T)$ when the current regime is i . Applying Itô's formula for regime-switching diffusions and invoking no-arbitrage:

$$\frac{\partial V_i}{\partial t} + (r - q)S \frac{\partial V_i}{\partial S} + \frac{1}{2}\sigma_i^2 S^2 \frac{\partial^2 V_i}{\partial S^2} - rV_i + \sum_{j=1}^N \lambda_{ij}(V_j - V_i) = 0, \quad i \in \mathcal{S}, \quad (11)$$

with $V_i(S, T) = \Phi(S)$ for all i .

The coupling term $\sum_j \lambda_{ij}(V_j - V_i)$ is the expected instantaneous change in option value due to regime transitions, discounted to present. It creates a *regime premium* embedded in the option price: the market pays more for an option in a low-vol regime than Black-Scholes predicts, because there is a positive probability of transitioning to the high-vol regime before expiry.

3.5 Two-Regime Closed-Form Approximation

For $N = 2$ with transition rates $\lambda_{12} = \lambda$ (low-to-high) and $\lambda_{21} = \mu$ (high-to-low), the stationary probabilities are $\pi_1 = \mu/(\lambda + \mu)$ and $\pi_2 = \lambda/(\lambda + \mu)$. The *effective mixture variance* is:

$$\bar{\sigma}^2 = \pi_1 \sigma_1^2 + \pi_2 \sigma_2^2 = \frac{\mu \sigma_1^2 + \lambda \sigma_2^2}{\lambda + \mu}. \quad (12)$$

Proposition 3.3 (ATM Implied Vol Under Regime Mixing). *To leading order in $(\sigma_2^2 - \sigma_1^2)^2/(\lambda + \mu)$, the at-the-money implied volatility $\hat{\sigma}$ satisfies:*

$$\hat{\sigma}^2 \approx \bar{\sigma}^2 - \frac{\lambda \mu (\sigma_2^2 - \sigma_1^2)^2}{4(\lambda + \mu)^3} \tau + \mathcal{O}((\lambda + \mu)^{-3}), \quad (13)$$

where $\tau = T - t$. The correction term is always **negative**, meaning the market systematically underprices ATM options relative to the stationary mixture variance when regimes switch rapidly.

Remark 3.4. Proposition 3.3 is the first formal justification for the practitioner observation that “vol of vol” suppresses ATM implied volatility. The faster the regime switching (large $\lambda + \mu$), the smaller the correction and the closer $\hat{\sigma}$ is to $\bar{\sigma}$.

3.6 Three-Regime Extension: Stress, Normal, and Crisis

For the empirically more realistic three-regime case that VRC uses in production ($\sigma_1 = 10\%$: calm, $\sigma_2 = 22\%$: normal, $\sigma_3 = 45\%$: crisis), the PDE system (11) with $N = 3$ is solved numerically by VRC’s pricing engine using a finite-difference scheme on a log-price grid with operator splitting for the coupling terms. The VRC generator

matrix is:

$$\Lambda^{\text{VRC}} = \begin{pmatrix} -(\lambda_{12} + \lambda_{13}) & \lambda_{12} & \lambda_{13} \\ \lambda_{21} & -(\lambda_{21} + \lambda_{23}) & \lambda_{23} \\ \lambda_{31} & \lambda_{32} & -(\lambda_{31} + \lambda_{32}) \end{pmatrix}. \quad (14)$$

VRC calibrates these transition rates to S&P 500 daily returns (2000–2024) using maximum likelihood on the hidden Markov model: $\lambda_{12} = 3.2$, $\lambda_{21} = 9.8$, $\lambda_{13} = 0.4$, $\lambda_{31} = 6.1$, $\lambda_{23} = 1.1$, $\lambda_{32} = 4.7$ (per year). The VRC-implied stationary distribution is approximately $(\pi_1, \pi_2, \pi_3) = (0.61, 0.32, 0.07)$, consistent with VRC’s observation that markets spend roughly 7% of time in crisis states and roughly 61% in identifiably calm conditions—the regime where VRC’s gamma scalping edge is largest and most reliable.

4. The Gamma Scalping P&L Identity Under Regime Switching

4.1 The VRC Delta-Hedged Portfolio

At each time t , the PRISM-managed VRC derivatives portfolio holds:

- +1 unit of the option (value $V_{X_t}(S_t, t)$), purchased at implied vol $\hat{\sigma}_0$ as recorded at VRC trade inception.
- $-\Delta_{X_t}(S_t, t)$ units of the spot-equivalent (executed via E-mini futures at VRC’s preferred hedge ratio).
- A cash position in the VRC margin account earning the risk-free rate r .

This portfolio is self-financing by construction: the delta-hedge leg is financed by borrowing/lending at r through VRC’s prime brokerage arrangement. All greeks are computed by VRC’s real-time pricing engine using the coupled PDE system (11) rather than flat Black-Scholes, ensuring that the delta reflects the VRC regime posterior.

4.2 Main P&L Identity

Theorem 4.1 (Regime-Switching Gamma Scalping P&L). *The discounted cumulative P&L of the continuously delta-hedged portfolio satisfies:*

$$e^{-rt}\Pi_t = \Pi_0 + \frac{1}{2} \int_0^t e^{-rs} S_s^2 \Gamma_{X_s}(S_s, s) \underbrace{(\sigma_{X_s}^2 - \widehat{\sigma}_{X_s}^2)}_{\text{regime variance spread}} ds + M_t^\Pi, \quad (15)$$

where $\widehat{\sigma}_i(S, t)$ is the Black-Scholes implied volatility of the option valued at V_i , and M_t^Π is a square-integrable $(\mathbb{P}, \mathcal{F})$ -martingale.

Proof. Apply Itô's product rule to $e^{-rt}\Pi_t$. By the generalized Itô formula for MRS-GBM:

$$dV_{X_t} = \left[\frac{\partial V_i}{\partial t} + \mu_i S_t \frac{\partial V_i}{\partial S} + \frac{1}{2} \sigma_i^2 S_t^2 \frac{\partial^2 V_i}{\partial S^2} \right] dt + \sigma_i S_t \Delta_i dW_t + \sum_{j \neq i} (V_j - V_i) dN_t^{ij}, \quad (16)$$

where N_t^{ij} is the counting process for transitions $i \rightarrow j$. Subtracting $\Delta_i dS_t = \Delta_i(\mu_i S_t dt + \sigma_i S_t dW_t)$ and invoking the pricing PDE (11) to cancel the drift components:

$$d\Pi_t = \frac{1}{2} \sigma_i^2 S_t^2 \Gamma_i dt - r(\Pi_t - V_i + \Delta_i S_t) dt + \sum_{j \neq i} (V_j - V_i) (dN_t^{ij} - \lambda_{ij} dt). \quad (17)$$

The dt term yields $\frac{1}{2} S_t^2 \Gamma_i (\sigma_i^2 - \widehat{\sigma}_i^2) dt + r\Pi_t dt$. The jump martingale and the Brownian integral form M_t^Π . Discounting and integrating gives (15). $\square$

4.3 Regime-by-Regime Decomposition

Proposition 4.2 (P&L Decomposition). *The expected discounted P&L decomposes as:*

$$\mathbb{E}[e^{-rT}\Pi_T - \Pi_0] = \frac{1}{2} \sum_{i=1}^N \underbrace{w_i(T)}_{\text{gamma weight}} \cdot \underbrace{(\sigma_i^2 - \widehat{\sigma}_i^2)}_{\text{variance edge}}, \quad (18)$$

where:

$$w_i(T) = \mathbb{E} \left[\int_0^T e^{-rs} S_s^2 \Gamma_i(S_s, s) \mathbf{1}_{\{X_s=i\}} ds \right] \quad (19)$$

is the expected discounted dollar-gamma exposure in regime i .

VRC economic interpretation. The total expected P&L is a gamma-weighted average of per-regime variance differences. At VRC, this formula drives the core entry decision: before initiating any long gamma position, the VRC derivatives analyst must verify that the right-hand side of (18) is positive under the current particle filter posterior $\hat{\pi}_t$. Even if the strategy loses money in low-vol regimes (because $\sigma_1 < \widehat{\sigma}_1$), it remains profitable overall if the high-vol regimes contribute enough

positive P&L and if those regimes have high gamma weights. This last point is particularly important at VRC: high-vol regimes tend to drive the underlying toward the option strike (at-the-money conditions, maximum gamma), which is precisely the mechanism that makes the weighted sum in (18) biased toward profitability.

4.4 The Effective Breakeven Volatility Surface

Define the *effective breakeven implied volatility* $\hat{\sigma}^*$ as the level at which $\mathbb{E}[e^{-rT}\Pi_T] = 0$:

$$\hat{\sigma}^{*2} = \frac{\sum_{i=1}^N w_i(T) \sigma_i^2}{\sum_{i=1}^N w_i(T)}. \quad (20)$$

The VRC position is entered only when the quoted $\hat{\sigma} < \hat{\sigma}^*$, verified by PRISM at the time of trade initiation. Critically, $\hat{\sigma}^*$ is a *gamma-weighted* average of regime volatilities, not the simple stationary average $\sigma_{RV,\infty}$. Because gamma is maximized at-the-money and the underlying is most likely to be near-the-money when volatility has recently spiked, $\hat{\sigma}^*$ is typically biased *toward* the high-volatility regimes. This is one of the most important practical insights from the VRC derivatives research: a trader who naively uses the rolling 30-day realized vol as their breakeven will systematically *underestimate* the true edge and under-size positions. PRISM corrects for this at the source by computing $\hat{\sigma}^*$ exactly via (20) using the current VRC particle filter posterior.

4.5 P&L Variance and Higher Moments

The variance of the discounted cumulative P&L is:

$$\text{Var}[e^{-rT}\Pi_T] = \underbrace{\int_0^T e^{-2rs} \sigma_s^4 S_s^4 \bar{\Gamma}_s^2 ds}_{\text{gamma convexity variance}} + \underbrace{\sum_{i < j} \nu_{ij} (V_i - V_j)^2}_{\text{regime-jump variance}}, \quad (21)$$

where ν_{ij} captures the frequency of $i \rightarrow j$ transitions. The second term—the *regime-jump variance*—is absent from single-regime models and is a source of P&L uncertainty that the VRC risk management framework specifically monitors. It is small when transition rates are low (infrequent regime shifts, typical of VRC’s calm regime) and large during rapid transitions (the crisis environment where VRC has already reduced position size via the entropy penalty (48)).

5. Futures Dynamics and Basis Arbitrage

5.1 Cost of Carry and the VRC Futures Pricing Standard

All VRC derivative positions that require a delta hedge use E-mini S&P 500 futures (ES), E-mini Nasdaq-100 futures (NQ), or the relevant single-stock futures where available. Let F_t be the futures price with delivery date $T_F > T$. The VRC pricing engine continuously monitors the theoretical cost-of-carry relationship:

$$F_t^{\text{fair}} = S_t e^{(r-q)(T_F-t)}, \quad (22)$$

where r is the continuously compounded risk-free rate (updated daily from the VRC FRED feed) and q is the dividend yield (sourced from VRC's Bloomberg terminal). This relationship holds exactly only in frictionless markets; in practice VRC observes persistent deviations that constitute the basis.

5.2 The Stochastic Basis

Definition 5.1 (Futures Basis). The traded futures basis is:

$$b_t = F_t - S_t e^{(r-q)(T_F-t)}. \quad (23)$$

Assumption 5.2 (Ornstein-Uhlenbeck Basis Dynamics). The basis b_t satisfies:

$$db_t = -\kappa b_t dt + \sigma_b dW_t^b, \quad (24)$$

where $\kappa > 0$, $\sigma_b > 0$, and $dW_t^b dW_t = \rho_b dt$ with $\rho_b \in (-1, 1)$. The VRC regime-switching basis extends this to $\kappa = \kappa_{X_t}$, $\sigma_b = \sigma_{b, X_t}$, capturing the empirically confirmed VRC observation that ES basis volatility is $2.1\times$ higher in the crisis regime than in the calm regime.

The stationary distribution of b_t under Assumption 5.2 is:

$$b_\infty \sim \mathcal{N}\left(0, \frac{\sigma_b^2}{2\kappa}\right), \quad (25)$$

with half-life of mean reversion $t_{1/2} = \ln 2 / \kappa$.

5.3 Futures Hedge Ratio and P&L

The delta hedge using futures requires h_t^F contracts where:

$$h_t^F = \Delta_{X_t}(S_t, t) \cdot e^{-(r-q)(T_F-t)}. \quad (26)$$

Proposition 5.3 (Futures-Hedged P&L Decomposition). *Under the futures hedge*

ratio (26), the discounted P&L is:

$$\begin{aligned} e^{-rt}\Pi_t^F &= \Pi_0 + \frac{1}{2} \int_0^t e^{-rs} S_s^2 \Gamma_{X_s} (\sigma_{X_s}^2 - \hat{\sigma}_{X_s}^2) ds \\ &\quad - \int_0^t e^{-rs} h_s^F db_s - \int_0^t e^{-rs} h_s^F (r - q) b_s ds + M_t^F. \end{aligned} \quad (27)$$

The third integral has expected value:

$$-\mathbb{E} \left[\int_0^t e^{-rs} h_s^F (r - q) b_s ds \right] = (r - q) \int_0^t e^{-rs} h_s^F \frac{\sigma_b^2}{2\kappa} (1 - e^{-2\kappa s})^{1/2} ds \geq 0, \quad (28)$$

whenever $(r - q) > 0$ **and the trader takes the correct sign on the basis position**, providing a strictly positive additional P&L stream.

The VRC basis alpha. Equation (28) is the central result of this section and one that the VRC derivatives desk exploits systematically: futures-based delta hedging generates a *free enhancement* to the gamma scalping P&L equal to the carry rate times the expected absolute basis. In the current carry environment (US equity dividend yields around 1.8% versus a risk-free rate of 5.25%), this enhancement is robustly positive on every long-gamma position VRC carries. In an inverted carry environment, VRC switches its basis positioning to capture mean-reversion income regardless of the carry sign, ensuring the third integral in (27) always works in VRC's favour.

5.4 Basis Roll Management

When the front-month futures contract approaches expiry, the basis undergoes a predictable compression toward zero (the futures price converges to spot at T_F). This roll effect creates both risk and opportunity. The roll P&L of a position that holds h_t^F futures contracts is:

$$\Pi^{\text{roll}} = h_{T_F^-}^F (F_{T_F^-} - S_{T_F^-}) = h_{T_F^-}^F b_{T_F^-}, \quad (29)$$

which is the basis at the roll date times the position size. The VRC derivatives desk rolls all futures positions Δ_{roll}^* days before expiry, where Δ_{roll}^* is solved by PRISM to minimize expected roll slippage:

$$\Delta_{\text{roll}}^* = \arg \min_{\Delta} \mathbb{E}[|b_{T_F - \Delta}|] \cdot c_{\text{roll}}, \quad (30)$$

with c_{roll} the round-trip cost of rolling one contract.

6. Arbitrage Theorems: Options, Futures, and Spot

6.1 The VRC Three-Asset No-Arbitrage Framework

VRC operates simultaneously in three markets for every gamma scalping position: the option market (where VRC is long), the futures market (where VRC hedges delta), and occasionally the spot market (for dividend adjustments and corporate actions). The formal no-arbitrage structure across these three markets is:

Definition 6.1 (VRC Admissible Portfolio). A portfolio $\Psi = (n_V, n_F, n_S, n_c)$ on the VRC derivatives book is *admissible* if it is self-financing, $\mathcal{F}_t$ -adapted, and its wealth process is bounded below $\mathbb{P}$ -a.s.

Theorem 6.2 (Three-Asset No-Arbitrage). *Under Assumption 3.1 and Assumption 5.2, the system $\{V_{X_t}, F_t, S_t\}$ admits no riskless arbitrage among admissible portfolios if and only if all three of the following hold:*

- (i) **Put-Call Parity** holds regime-by-regime: $C_i(S, t) - P_i(S, t) = Se^{-q\tau} - Ke^{-r\tau}$ for all $i \in \mathcal{S}$.
- (ii) **Futures-Spot Consistency**: $\mathbb{E}^\mathbb{Q}[F_{T_F} \mid \mathcal{F}_t] = S_t e^{(r-q)(T_F-t)} + b_t e^{-\kappa(T_F-t)}$.
- (iii) **PDE Validity**: The pricing PDE system (11) holds for all $i \in \mathcal{S}$ and $(S, t) \in \mathbb{R}_{>0} \times [0, T]$.

6.2 The PRISM Arbitrage Condition

Theorem 6.3 (PRISM Structural Arbitrage). *Suppose (10) and the ergodic convergence (9) hold. The gamma scalping portfolio generates strictly positive expected discounted P&L if and only if:*

$$\sum_{i=1}^N \pi_i \bar{\Gamma}_i (\sigma_i^2 - \hat{\sigma}_i^2) > 0, \quad (31)$$

where $\bar{\Gamma}_i = \mathbb{E}[S_t^2 \Gamma_i(S_t, t) \mid X_t = i]$ is the regime-conditional expected dollar gamma. The annualized arbitrage P&L rate is:

$$\alpha^{arb} = \frac{1}{2} \sum_{i=1}^N \pi_i \bar{\Gamma}_i (\sigma_i^2 - \hat{\sigma}_i^2) > 0. \quad (32)$$

Corollary 6.4 (Basis-Enhanced Arbitrage). *The total P&L of the futures-hedged*

strategy exceeds the spot-hedged strategy by the basis alpha:

$$\mathbb{E}[\Pi_T^F] - \mathbb{E}[\Pi_T^S] = (r - q) \int_0^T \mathbb{E}\left[h_s^F \frac{\sigma_b^2}{2\kappa_{X_s}}\right] ds - c_\Delta^F \cdot \mathbb{E}[N_T^F] + c_\Delta^S \cdot \mathbb{E}[N_T^S], \quad (33)$$

where $c_\Delta^F < c_\Delta^S$ is the per-unit transaction cost for futures versus spot (at VRC, $c_\Delta^F \approx 2.5$ bps versus $c_\Delta^S \approx 9$ bps), and N_T^F, N_T^S are the number of rebalances under each hedging approach. For any liquid futures market in VRC's coverage universe, the right-hand side is **positive** by a margin that VRC estimates at 15–25 bps per year of annualized notional, depending on the carry environment and basis regime.

7. Optimal Hedging Bandwidth: HJB Formulation

7.1 VRC Discrete Rebalancing with Transaction Costs

Continuous rebalancing is theoretically optimal but operationally impossible: every trade at VRC incurs prime brokerage commissions, exchange fees, and bid-ask spread. The VRC derivatives desk rebalances only when the delta mismatch δ_t exceeds the PRISM-computed threshold $\hat{\eta}_t^*$. Each rebalance of size $|dn|$ incurs the VRC effective cost $c_{\text{eff}} \cdot S_t \cdot |dn|$ from equation (58).

7.2 The HJB Equation

Define the value function:

$$J(S, \delta, b, t, i) = \sup_{\eta \geq 0} \mathbb{E}[\Pi_T^\eta \mid S_t = S, \delta_t = \delta, b_t = b, X_t = i]. \quad (34)$$

In the *no-rebalancing region* $\{|\delta| < \eta\}$, the HJB equation is:

$$\begin{aligned} \frac{\partial J}{\partial t} + (r - q)S \frac{\partial J}{\partial S} + \frac{\sigma_i^2 S^2}{2} \frac{\partial^2 J}{\partial S \partial S} - \kappa b \frac{\partial J}{\partial b} + \frac{\sigma_b^2}{2} \frac{\partial^2 J}{\partial b \partial b} \\ + \sigma_i^2 S^2 \Gamma_i \frac{\partial J}{\partial \delta} + \frac{\sigma_i^2 S^2 \Gamma_i^2}{2} \frac{\partial^2 J}{\partial \delta \partial \delta} + \sum_j \lambda_{ij} (J_j - J_i) = -\frac{1}{2} \sigma_i^2 S^2 \Gamma_i (\sigma_i^2 - \hat{\sigma}_i^2), \end{aligned} \quad (35)$$

with the *smooth-pasting* and *value-matching* conditions at $|\delta| = \eta^*$:

$$J|_{|\delta|=\eta^*} = J|_{\delta=0} - cS|\eta^*|, \quad \left. \frac{\partial J}{\partial \delta} \right|_{|\delta|=\eta^*} = -cS \cdot \text{sgn}(\delta). \quad (36)$$

7.3 Closed-Form Optimal Bandwidth

Theorem 7.1 (Regime-Adjusted Optimal Bandwidth). *In the limit $c \rightarrow 0$, $\lambda_{ij} \rightarrow 0$, the optimal delta-mismatch threshold in regime i is:*

$$\eta_i^* \approx \left(\frac{3c\sigma_i}{\Gamma_i S} \right)^{1/3} \cdot \left[1 + \underbrace{\frac{\sigma_i^2 - \hat{\sigma}_i^2}{3\sigma_i^2}}_{\text{regime correction}} \right] + \mathcal{O}(c^{2/3}), \quad (37)$$

where the leading factor $(3c\sigma_i/\Gamma_i S)^{1/3}$ is the Whalley-Wilmott bandwidth [Whalley and Wilmott, 1997] and the bracketed correction reflects the regime-specific variance spread.

Remark 7.2. When $\sigma_i^2 > \hat{\sigma}_i^2$ (realized exceeds implied in regime i , the profitable case that VRC enters), the correction factor exceeds 1, meaning the PRISM-prescribed bandwidth is **wider** than the neutral Whalley-Wilmott level. The VRC economic intuition is precise: gamma income partially compensates for the cost of rebalancing, so the urgency to rebalance is reduced. VRC analysts should not override PRISM's bandwidth to tighten it during winning periods; this is the most common behavioral error the VRC risk committee has observed in manual gamma scalping, and it costs approximately 8–12 bps/year in unnecessary transaction costs.

7.4 Regime-Switching Bandwidth Schedule

The bandwidth adapts in real time to the estimated regime:

$$\hat{\eta}_t^* = \sum_{i=1}^N \hat{\pi}_{i,t} \eta_i^*, \quad (38)$$

where $\hat{\pi}_{i,t}$ is the VRC particle filter posterior probability of regime i at time t (Section 9). This adaptive bandwidth is computed by PRISM and broadcast to the VRC derivatives desk every morning before market open and updated intraday whenever the particle filter detects a meaningful shift in the regime posterior (defined as a change of > 0.10 in any $\hat{\pi}_{i,t}$). The bandwidth automatically tightens during regime ambiguity (when the VRC entropy $H(\hat{\pi}_t)$ is high) and loosens when the regime is clearly identified—systematically enforcing trading discipline that VRC would otherwise have to impose manually.

8. P&L Distribution: Fokker-Planck and Kelly Sizing

8.1 Fokker-Planck Equation for Cumulative P&L

Let $p_i(x, t) = \frac{\partial}{\partial x} \mathbb{P}(\Pi_t \leq x, X_t = i)$ be the joint density of P&L and regime. From the SDE for Π_t implied by Theorem 4.1, the Fokker-Planck equation is:

$$\frac{\partial p_i}{\partial t} = -\frac{\partial}{\partial x} [\mu_i^{\text{P\&L}} p_i] + \frac{1}{2} \frac{\partial^2}{\partial x^2} [(\sigma_i^{\text{P\&L}})^2 p_i] + \sum_j \lambda_{ji} p_j - \sum_j \lambda_{ij} p_i, \quad (39)$$

where the regime-conditional P&L drift and diffusion are:

$$\mu_i^{\text{P\&L}}(t) = \frac{1}{2} S_t^2 \Gamma_i (\sigma_i^2 - \hat{\sigma}_i^2), \quad (40)$$

$$(\sigma_i^{\text{P\&L}})^2(t) = \sigma_i^4 S_t^4 \Gamma_i^2. \quad (41)$$

The marginal P&L density is $p(x, t) = \sum_i p_i(x, t)$.

8.2 Analytical Moments: What VRC Expects from Each Trade

Under the long-run ergodic approximation, the VRC derivatives desk can form closed-form expectations for every P&L moment before initiating a position. These are not post-hoc rationalizations; they are pre-trade targets that PRISM computes at inception and that the VRC thesis integrity monitor tracks throughout the position's life.

Mean:

$$\mathbb{E}[\Pi_T] = \frac{T}{2} \sum_{i=1}^N \pi_i \bar{\Gamma}_i S^2 (\sigma_i^2 - \hat{\sigma}_i^2) = \alpha^{\text{arb}} \cdot T. \quad (42)$$

Variance:

$$\text{Var}[\Pi_T] = T \sum_{i=1}^N \pi_i \sigma_i^4 S^4 \bar{\Gamma}_i^2 + T \sum_{i < j} \nu_{ij} (V_i - V_j)^2 \equiv T \cdot (\mathbf{v}^{\text{diff}} + \mathbf{v}^{\text{jump}}). \quad (43)$$

Skewness (third central moment, leading order):

$$\mu_3[\Pi_T] \approx T \sum_{i=1}^N \pi_i \sigma_i^6 S^6 \Gamma_i^3 > 0, \quad (44)$$

confirming that VRC gamma scalping P&L distributions are *right-skewed*: there are

occasional large positive days when vol spikes and gamma income is large, and a steady small drain of theta and transaction costs on quiet days. This is a feature, not a bug, of the VRC strategy: the right-skew means that the strategy’s worst days are bounded (theta drain) while the best days are unbounded (gamma windfall). The VRC investment committee tracks the realized skewness of each position against the theoretical prediction (44) as a model health metric.

8.3 The VRC PRISM Sharpe Ratio: The Pre-Trade Hurdle

Before any position is opened, PRISM computes the expected Sharpe ratio of the proposed trade using the current regime posterior. The VRC investment committee requires a minimum pre-trade Sharpe of 1.2 (after transaction costs) as one of the four entry conditions in the VRC protocol (Section 14). The analytical Sharpe formula is:

$$\text{SR} = \frac{\alpha^{\text{arb}}}{\sqrt{\mathbf{v}^{\text{diff}} + \mathbf{v}^{\text{jump}}}} = \frac{\frac{1}{2} \sum_i \pi_i \bar{\Gamma}_i S^2 (\sigma_i^2 - \hat{\sigma}_i^2)}{\sqrt{\sum_i \pi_i \sigma_i^4 S^4 \bar{\Gamma}_i^2 + \sum_{i < j} \nu_{ij} (V_i - V_j)^2}}. \quad (45)$$

8.4 VRC Kelly Criterion and Fractional Sizing

Position sizing at VRC is not discretionary. PRISM computes the Kelly-optimal fraction of VRC’s derivatives capital allocation at trade inception and after every particle filter update. Under log-utility maximization, the Kelly-optimal fraction is:

$$f^* = \frac{\mu_{\text{avg}}^{\text{P\&L}}}{(\sigma_{\text{avg}}^{\text{P\&L}})^2} = \text{SR} \cdot \frac{1}{\sigma_{\text{avg}}^{\text{P\&L}}}. \quad (46)$$

VRC applies a *fractional Kelly* factor of 1/2 as standard policy, consistent with the VRC investment committee’s standing decision to account for model estimation error in all systematic strategies—a policy that applies equally to AEGIS position sizing [Verma, 2026a] and to PRISM:

$$f^{\text{Kelly}} = \frac{1}{2} f^*. \quad (47)$$

The *VRC regime-uncertainty penalty* modifies the Kelly fraction further, automatically reducing the VRC derivatives allocation when the particle filter is uncertain about the current regime:

$$f_{\text{VRC}}^{\text{adj}} = f^{\text{Kelly}} \cdot \exp(-\lambda_{\text{unc}} \cdot H(\hat{\boldsymbol{\pi}}_t)), \quad (48)$$

where $H(\hat{\boldsymbol{\pi}}_t) = -\sum_i \hat{\pi}_{i,t} \ln \hat{\pi}_{i,t}$ is the Shannon entropy of the VRC regime posterior. When the VRC filter is confident ($H \rightarrow 0$): $f_{\text{VRC}}^{\text{adj}} \rightarrow f^{\text{Kelly}}$, the full half-Kelly alloca-

tion is deployed. When regime is maximally uncertain ($H = \ln N$): the VRC allocation shrinks by factor $e^{-\lambda_{\text{unc}} \ln N} = N^{-\lambda_{\text{unc}}}$. At VRC's calibrated $N = 3$ regimes and $\lambda_{\text{unc}} = 0.5$, this is a 42% automatic reduction in position size—a meaningful, systematic de-risking that happens without any analyst intervention. The VRC investment committee sets λ_{unc} annually; the current value of 0.5 reflects the committee's view that regime uncertainty is moderately costly relative to the cost of missing a good trade.

8.5 VRC Tail Risk Management and Value-at-Risk

The VRC risk management framework requires that every derivatives position report a daily Cornish-Fisher-adjusted VaR to the VRC risk committee. The unadjusted VaR at confidence level α is:

$$\text{VaR}_\alpha^{\text{VRC}}(\Pi_T) = -\mu_{\text{P\&L}}T + z_{1-\alpha}\sqrt{\text{Var}[\Pi_T]}, \quad (49)$$

where $z_{1-\alpha}$ is the $(1 - \alpha)$ -quantile of the standard normal. Due to the right-skewness of the VRC P&L distribution (44), the normal approximation understates the true VaR at high confidence levels; PRISM corrects for this using the Cornish-Fisher expansion, which the VRC risk committee has mandated for all options positions:

$$z_{1-\alpha}^{\text{CF}} = z_{1-\alpha} + \frac{\hat{\gamma}_1}{6}(z_{1-\alpha}^2 - 1) + \frac{\hat{\gamma}_2}{24}(z_{1-\alpha}^3 - 3z_{1-\alpha}) - \frac{\hat{\gamma}_1^2}{36}(2z_{1-\alpha}^3 - 5z_{1-\alpha}), \quad (50)$$

where $\hat{\gamma}_1$ and $\hat{\gamma}_2$ are the estimated skewness and excess kurtosis of the P&L distribution.

9. Real-Time Regime Detection: The VRC Particle Filter

9.1 Why VRC Needs a Real-Time Regime Filter

Every trading decision PRISM makes—the bandwidth $\hat{\eta}_t^*$, the Kelly fraction $f_{\text{VRC}}^{\text{adj}}$, the VRP z-score, the arbitrage condition check—depends on the current regime posterior $\hat{\pi}_t$. This is not an optional analytical refinement; it is the operational core of the VRC derivatives book. A stale or inaccurate regime estimate is the single largest source of edge erosion in gamma scalping, as VRC's own historical P&L attribution confirms: the bottom decile of VRC trade P&L is dominated by positions entered when the particle filter's regime confidence (measured by $1 - H / \ln N$) was below 0.6. The VRC filter solves the problem of estimating X_t from observables. The VRC

derivatives desk does not observe the regime directly; it observes prices, returns, and the options surface. PRISM uses a Sequential Importance Resampling (SIR) particle filter augmented with three signal sources that represent VRC's proprietary regime-detection innovations relative to standard academic implementations: (i) the live SPX and single-stock options chain (the VRC options surface likelihood, described below), (ii) Garman-Klass realized vol estimators with multiple lookback windows (5, 10, 22 days), and (iii) the VIX futures term structure slope.

9.2 The VRC Multi-Signal Likelihood Function

The VRC filter's advantage over a pure-returns filter lies in its composite likelihood, which fuses three independent data streams. The likelihood of observing VRC's data at epoch t_k given regime i is:

$$\ell_k^{(i)} = \underbrace{\phi\left(\frac{r_k - (m_i - \frac{1}{2}\sigma_i^2)\Delta}{\sigma_i\sqrt{\Delta}}\right)}_{\text{return likelihood}} \cdot \underbrace{f_\chi\left(\frac{(M-1)\hat{\sigma}_k^2}{\sigma_i^2}; M-1\right)}_{\text{rolling vol likelihood}} \cdot \underbrace{g_i(\text{ATM}_k, \text{skew}_k)}_{\text{options surface signal}}, \quad (51)$$

where ϕ is the standard normal pdf, $r_k = \ln(S_{t_k}/S_{t_{k-1}})$ is the log-return, $\hat{\sigma}_k^2$ is the realized variance over M preceding periods estimated using the Garman-Klass estimator [Garman and Klass, 1980], $f_\chi(\cdot; M-1)$ is the chi-squared density, and $g_i(\cdot)$ is the options-based likelihood described below.

VRC options-surface likelihood. The live options chain that VRC monitors in real time encodes forward-looking volatility information that is not yet in realized returns. For each regime i , PRISM computes the VRC model ATM implied vol $\hat{\sigma}_i^{\text{VRC}}$ and model skew sk_i^{VRC} from the coupled PDE system (11) evaluated at the current VRC regime parameters. The options likelihood is:

$$g_i(\text{ATM}_k, \text{skew}_k) \propto \exp\left(-\frac{(\text{ATM}_k - \hat{\sigma}_i^{\text{VRC}})^2}{2\omega_{\text{ATM}}^2} - \frac{(\text{skew}_k - \text{sk}_i^{\text{VRC}})^2}{2\omega_{\text{sk}}^2}\right), \quad (52)$$

where $\omega_{\text{ATM}} = 0.8\%$ and $\omega_{\text{sk}} = 0.3\%$ are VRC-calibrated noise parameters (re-estimated quarterly using VRC's options data history). This options channel is VRC's key competitive advantage in regime detection: it allows PRISM to identify regime shifts *before* they appear in realized price action, because the options market prices in upcoming volatility before it materializes in returns. VRC's own testing on the 2000–2024 S&P 500 data confirms an average 1.8-day lead of the options signal over the pure-returns filter in detecting regime transitions (Table 4).

9.3 The VRC SIR Algorithm

Algorithm 1 VRC PRISM Multi-Signal Particle Filter (runs at VRC market open and on intraday triggers)

- 1: **Initialize:** Sample $x_0^{(m)} \sim \text{Uniform}(\mathcal{S})$, set $w_0^{(m)} = 1/M$ for all $m = 1, \dots, M = 20,000$.
 - 2: **for** each VRC observation epoch t_k (daily close + intraday triggers) **do**
 - 3: **Propagate:** $x_{t_k}^{(m)} \sim P(\Delta t)[x_{t_{k-1}}^{(m)}, \cdot]$ where $P(\Delta t) = e^{\Lambda^{\text{VRC}} \Delta t}$.
 - 4: **Compute VRC likelihood:** $\ell_k^{(m)}$ via (51) using VRC live return, GK realized vol, and options chain data.
 - 5: **Update weights:** $\tilde{w}_{t_k}^{(m)} = w_{t_{k-1}}^{(m)} \cdot \ell_k^{(m)}$.
 - 6: **Normalize:** $w_{t_k}^{(m)} = \tilde{w}_{t_k}^{(m)} / \sum_{m'} \tilde{w}_{t_k}^{(m')}$.
 - 7: **Compute VRC posterior:** $\hat{\pi}_{i,t_k}^{\text{VRC}} = \sum_m w_{t_k}^{(m)} \mathbf{1}_{\{x_{t_k}^{(m)} = i\}}$.
 - 8: **Compute VRC entropy:** $H_{t_k}^{\text{VRC}} = - \sum_i \hat{\pi}_{i,t_k}^{\text{VRC}} \ln \hat{\pi}_{i,t_k}^{\text{VRC}}$.
 - 9: **Resample** if $N_{\text{eff}} = 1 / \sum_m (w_{t_k}^{(m)})^2 < M/2$ using systematic resampling.
 - 10: **Broadcast to VRC desk:** Updated $\hat{\pi}_{t_k}^{\text{VRC}}$, $\hat{\eta}_{t_k}^*$, and $f_{\text{VRC}}^{\text{adj}}$ via (38) and (48).
 - 11: **end for**
-

Theorem 9.1 (Particle Filter Convergence). *Under standard boundedness conditions on $\ell_k^{(m)}$, the SIR estimate $\hat{\pi}_{t_k}$ satisfies:*

$$\sup_{k \leq T/\Delta t} \|\hat{\pi}_{t_k} - \pi_{t_k}^{\text{true}}\|_1 \xrightarrow{M \rightarrow \infty} 0 \quad \mathbb{P}\text{-a.s.}, \quad (53)$$

with L^2 convergence rate $\mathcal{O}(M^{-1/2})$ [Del Moral, 2004]. At VRC's production setting of $M = 20,000$ particles, the expected L^1 error in $\hat{\pi}^{\text{VRC}}$ is below 0.054 (Table 4), which VRC's back-testing confirms is well within the precision needed for reliable bandwidth and Kelly calculations.

10. Variance Risk Premium Forecasting

10.1 The VRP as the Core VRC Entry Signal

The variance risk premium $\text{VRP}_t = \hat{\sigma}_t^2 - \mathbb{E}[\sigma_{\text{RV},t:T}^2]$ is the number VRC watches most closely. A large, positive, regime-adjusted VRP means the market is overpaying for volatility insurance relative to what VRC's model estimates will actually be realized. VRC buys that insurance, delta-hedges it via PRISM, and harvests the difference. The VRP z-score (56) is the operational translation of this idea into a binary entry

signal.

10.2 The VRC HAR-RV-Regime Model

Simple rolling-window realized vol is a poor estimator of the VRP because it conflates returns from different regimes. VRC's proprietary improvement, the HAR-RV-Regime model, weights historical realized variance by the current regime posterior, yielding a regime-adjusted expected RV forecast that is more accurate than any single-lookback estimator. Building on the heterogeneous autoregressive model of Corsi [2009], the VRC HAR-RV-Regime equation is:

$$\sigma_{RV,t+h} = \beta_0 + \sum_{i=1}^N \hat{\pi}_{i,t} \left(\beta_d^{(i)} \sigma_{RV,d} + \beta_w^{(i)} \sigma_{RV,w} + \beta_m^{(i)} \sigma_{RV,m} \right) + \epsilon_{t+h}, \quad (54)$$

where $\sigma_{RV,d}$, $\sigma_{RV,w}$, $\sigma_{RV,m}$ are the realized variance computed over the past 1, 5, and 22 days, and $\hat{\pi}_{i,t}^{VRC}$ is the VRC regime posterior. The regime-conditional coefficients $\beta^{(i)}$ are re-estimated quarterly by VRC using weighted OLS on the full VRC historical data set, with observation weights $\hat{\pi}_{i,t}^{VRC}$.

10.3 VRC VRP Regime Decomposition

The VRC VRP can be decomposed into regime-conditional components:

$$VRP_t^{VRC} = \sum_{i=1}^N \hat{\pi}_{i,t}^{VRC} VRP_t^{(i)}, \quad VRP_t^{(i)} = \hat{\sigma}_t^2 - \beta_0^{(i)} - \beta_d^{(i)} \sigma_{RV,d} - \beta_w^{(i)} \sigma_{RV,w} - \beta_m^{(i)} \sigma_{RV,m}. \quad (55)$$

This decomposition confirms what VRC's own trading experience reflects (Table 2): the VRP is large and reliable in VRC's calm regime, meaningful in the normal regime, and frequently negative in the crisis regime. The VRC derivatives desk therefore concentrates its gamma scalping activity in calm and normal regimes and reduces or exits all long gamma positions when the VRC filter assigns crisis probability above 60%.

10.4 VRC Entry Signal: The VRP Z-Score

The VRC derivatives team enters a long gamma position when PRISM's normalized VRP signal exceeds the VRC minimum threshold $z^* = 1.0$:

$$z_t^{VRP} = \frac{VRP_t - \mu_{X_t}^{VRP}}{\sigma_{X_t}^{VRP}} > z^*, \quad (56)$$

where μ_i^{VRP} and σ_i^{VRP} are the VRC regime-conditional mean and standard deviation of the VRP from the VRC HAR-regime model. The VRC threshold $z^* = 1.0$ corresponds to a VRP in the top 16% of regime-adjusted historical observations—a level that VRC’s back-testing shows produces a favourable hit rate without being so restrictive that it generates fewer than two new positions per month across the VRC coverage universe.

11. Volatility Surface Dynamics and Skew Management

11.1 How VRC Reads the Regime-Switching Vol Surface

The VRC derivatives desk monitors the SPX implied volatility surface in real time through Bloomberg and VRC’s own options analytics layer. The MRSGBM model predicts specific surface shapes in each regime, and deviations from those predictions are themselves informative signals. The ATM term structure of implied vol reflects the blend of VRC’s regime volatilities weighted by occupancy probabilities over each tenor; the skew reflects the asymmetry in the distribution of returns induced by the regime transition probabilities embedded in Λ^{VRC} .

Under the VRC two-regime MRSGBM, the risk-neutral probability of a large down-move (which corresponds to a transition to the high-vol regime) exceeds the log-normal probability, generating a *negative skew* consistent with the equity market’s well-documented left tail. The VRC formula for the slope of the ATM implied vol skew is:

$$\left. \frac{\partial \hat{\sigma}}{\partial \Delta} \right|_{\Delta=0.5} \approx -\frac{\lambda_{12}(\sigma_2 - \sigma_1)}{(\lambda_{12} + \lambda_{21})\sigma_{\text{eff}}} \cdot \sqrt{\frac{2\tau}{\pi}}, \quad (57)$$

where $\sigma_{\text{eff}} = \sqrt{\hat{\sigma}^2}$ is the effective ATM vol. This expression shows that the VRC-predicted skew is proportional to the transition rate from low-to-high vol and to the vol spread, consistent with VRC’s observation that SPX skew steepens in the 2–3 weeks before elevated-volatility episodes. When the observed market skew diverges materially from equation (57)’s prediction, it signals either a regime mispricing or a non-MRSGBM shock, both of which are reviewed by the VRC derivatives team before any new position is opened.

11.2 VRC Vega Hedging and Skew Harvesting

Long gamma positions are inherently long vega. The VRC derivatives book manages vega exposure through two systematic channels:

(i) **VRC vega bucketing.** Every VRC option position is decomposed into vega exposure by expiry tenor $\{\mathcal{V}_{T_1}, \dots, \mathcal{V}_{T_K}\}$ in PRISM’s Greeks engine. VRC hedges residual vega using a combination of shorter- and longer-dated options to maintain near-zero net vega at each tenor while preserving the gamma position. The VRC vega concentration limit (3% of AUM net vega) is enforced at the portfolio level by the PRISM risk monitor.

(ii) **VRC skew harvesting.** When the observed SPX skew exceeds the VRC model-predicted skew from (57) by more than 1.5 vol points at the 25-delta put strike, the market is paying excess downside protection premium relative to the VRC MRSGBM calibration. VRC monetizes this by selling put spreads (long the 25-delta put, short the 10-delta put) while maintaining the core long gamma position, capturing the excess skew premium with defined downside. This skew overlay typically contributes 8–14 bps of additional annualized P&L to the VRC derivatives book in normal-regime environments.

12. VRC Microstructure and Execution Framework

12.1 VRC Transaction Cost Model

Every parameter in the PRISM framework that involves a cost—the Kelly fraction, the optimal bandwidth, the roll timing, the VWAP threshold—is calibrated to the *actual* VRC transaction cost structure rather than a generic industry estimate. The VRC effective round-trip cost for a delta hedge of size $|dn|$ is:

$$c_{\text{eff}} = c_{\text{spread}} + c_{\text{impact}}(|dn|) + c_{\text{borrow}}, \quad (58)$$

where c_{spread} is VRC’s observed bid-ask half-spread (2.5 bps for E-mini futures, 4–8 bps for liquid single-stock options), $c_{\text{impact}}(|dn|) = \kappa_I |dn|/V_{30d}$ is the square-root market impact calibrated quarterly by VRC to V_{30d} (the 30-day average daily volume of the underlying), and c_{borrow} is VRC’s current securities lending rate for short delta positions. These inputs are updated in PRISM’s parameter database every Monday morning.

12.2 VRC VWAP-Optimal Delta Hedge Execution

When the required delta adjustment $|dn|$ is large relative to market liquidity (more than 1% of ADV), the VRC execution desk uses a VWAP schedule computed by PRISM to minimize market impact. The VRC optimal execution rate at time s

within the execution window $[t, t + T_{\text{exec}}]$ is:

$$\dot{n}_s^* = \frac{|dn|}{T_{\text{exec}}} + \lambda_{\text{pred}} \alpha_s, \quad (59)$$

where α_s is VRC's short-term alpha signal derived from E-mini order flow imbalance, and λ_{pred} is VRC's predictive trading parameter (calibrated monthly to minimise implementation shortfall). The VRC execution desk applies VWAP scheduling on any delta adjustment exceeding 1% of the underlying's 30-day ADV; smaller adjustments are executed at market.

12.3 VRC Options Liquidity Screen

Not all options in the VRC coverage universe meet the VRC liquidity standard. Before PRISM is permitted to enter a position on any name, the VRC liquidity screen must pass all four criteria:

- ATM bid-ask spread in vol terms < 50 bps (VRC threshold; the VRC investment committee reviews names between 35–50 bps quarterly and may restrict them to index-only positions).
- Open interest > 500 contracts on the target expiry.
- Average daily volume in the underlying > \$500M (ensuring the E-mini hedge market is deep).
- Options average daily volume > 1,000 contracts on the target expiry series.

Options failing any of these criteria are removed from the PRISM eligible universe and flagged to the VRC derivatives team. The VRC liquidity screen is re-run every Monday morning and after any earnings announcement or index reconstitution event.

13. VRC Numerical Results and Simulation Backtest

13.1 VRC Production Calibration

The parameters below are VRC's current production calibration of PRISM, derived from maximum-likelihood estimation on S&P 500 daily returns and SPX options data spanning January 2000 through December 2024. All VRC trading decisions described in Section 14 use these parameters unless a scheduled quarterly re-estimation produces a materially different fit (defined as a change of more than 15% in any σ_i or λ_{ij}).

Table 1: VRC PRISM Production Calibration: S&P 500 (2000–2024)

Regime	σ_i	π_i	VRC Description	Mean Duration	ATM IV
1 (VRC Calm)	10.2%	0.61	Low vol, trending market	18.4 days	13.8%
2 (VRC Normal)	22.1%	0.32	Mid vol, range-bound	8.7 days	22.9%
3 (VRC Crisis)	44.8%	0.07	High vol, stress event	4.1 days	38.4%

VRC additional production parameters: risk-free rate $r = 5.25\%$ (current Fed funds effective rate, updated daily), dividend yield $q = 1.8\%$ (SPX trailing 12-month yield), $K = S_0$ (ATM options, rolled monthly), $T = 0.25$ yr (3-month option expiry, VRC’s preferred tenor), basis mean-reversion $\kappa = 12 \text{ yr}^{-1}$, basis volatility $\sigma_b = 0.08\%$, basis-spot correlation $\rho_b = -0.12$, VRC futures transaction cost $c^F = 2.5$ bps, VRC spot transaction cost $c^S = 9$ bps, particle count $M = 20,000$, VRC uncertainty aversion $\lambda_{\text{unc}} = 0.5$.

13.2 VRP Statistics by Regime

Table 2: Variance Risk Premium by Regime: S&P 500 (2000–2024)

Regime	Mean VRP (ann. vol pts)	Std Dev	Sharpe of VRP	% Positive
1 (Calm)	+3.6%	4.1%	+0.88	72%
2 (Normal)	+1.4%	6.3%	+0.22	58%
3 (Crisis)	−5.2%	12.8%	−0.41	34%
Mixed	+2.3%	5.9%	+0.39	64%

These VRC simulation results confirm the core thesis: the VRP is large and reliable in the VRC calm regime (Sharpe 0.88, positive 72% of months), smaller but still positive in the normal regime, and frequently *negative* in crisis—exactly the pattern that drives VRC’s regime-conditional entry and exit rules. The VRC derivatives team does not fight the crisis-regime VRP inversion; PRISM’s entropy exit condition ensures VRC is out before the inversion is fully confirmed.

13.3 VRC Monte Carlo Simulation: Three Strategy Variants (10,000 Paths)

Table 3: VRC PRISM Strategy Performance: Monte Carlo Simulation (10,000 Paths, 3-Month Horizon)

Metric	Spot Hedge	Futures Hedge	VRC PRISM (Adaptive)
Annualized Mean P&L (bps of notional)	74.1	91.8	108.3
Annualized Std Dev P&L	55.2	52.4	44.7
Sharpe Ratio	1.34	1.75	2.42
Transaction Costs (bps/yr)	22.4	5.8	5.8
Net Mean P&L	51.7	86.0	102.5
Max Drawdown (bps)	−68.2	−52.6	−39.4
Skewness of P&L	+0.82	+0.91	+1.23
99% CVaR	−24.1	−18.7	−14.3
Prob. of Profit (single path)	68.4%	73.2%	79.6%
VRC Kelly Fraction (half-Kelly)	0.61	0.83	0.96

The VRC PRISM adaptive framework delivers a Sharpe ratio of 2.42 versus 1.34 for the naïve spot-hedged version—an 81% improvement that the VRC investment committee considers the clearest quantitative validation of the regime-switching approach. The three columns represent a natural hierarchy: spot hedging is what a textbook trader does; futures hedging is what a cost-aware trader does; VRC PRISM adaptive is what the VRC derivatives desk does. Three improvements drive the gap: (1) futures transaction cost savings (22.4 bps/yr reduced to 5.8 bps/yr); (2) regime-adaptive bandwidth that avoids over-trading in low-edge environments, reducing vol P&L standard deviation from 55.2 to 44.7; and (3) entropy-based position reduction during regime transitions that cuts max drawdown from 68.2 to 39.4 bps. The right-skewness improvement (0.82 to 1.23) is also meaningful: it confirms that VRC PRISM’s regime awareness correctly concentrates position size in high-edge environments (where the large positive gamma days occur) while reducing exposure during uncertain periods.

13.4 Regime Detection Performance

Table 4: VRC Particle Filter: Regime Detection Accuracy ($M = 20,000$ particles, S&P 500 2000–2024)

Metric	VRC Regime 1	VRC Regime 2	VRC Regime 3	Overall
Correct identification rate	93.1%	89.4%	86.2%	91.3%
Mean detection lag (trading days)	1.4	2.0	2.6	1.7
False transition rate	2.8%	3.1%	1.9%	2.7%
L^1 error in $\hat{\pi}^{\text{VRC}}$	0.048	0.061	0.074	0.054
VRC options signal contribution	+4.2%	+3.8%	+6.1%	+4.4%

The VRC options surface signal (final row) improves identification accuracy by 4–6% versus a returns-only filter across all regimes, with the largest improvement in the crisis regime where the options market prices in stress before it fully materialises in realised returns. This is precisely the regime where detection latency is most costly—a 2.6-day lag in entering the crisis regime can mean holding an incorrectly-sized long gamma position through the most violent vol spike of the episode. The VRC entropy penalty (48) automatically compensates for this latency by reducing the VRC position size during the ambiguous transition window.

13.5 Sensitivity Analysis

Table 5: VRC PRISM Sharpe Ratio Sensitivity to Key Production Parameters

VRC Parameter	−30%	−15%	VRC Base	+15%	+30%	Elasticity
Transaction cost c	2.71	2.57	2.42	2.28	2.14	−0.58
Regime switching rate $\bar{\lambda}$	2.19	2.31	2.42	2.53	2.61	+0.39
Vol spread $\sigma_3 - \sigma_1$	2.02	2.22	2.42	2.61	2.80	+0.78
Particle count M	2.30	2.38	2.42	2.44	2.45	+0.09
VRC Kelly fraction	1.89	2.18	2.42	2.58	2.64	+0.46

The VRC sensitivity analysis yields three actionable conclusions. **First**, transaction costs are VRC’s most controllable risk lever: a 30% reduction in execution costs (e.g., through tighter prime brokerage terms or improved execution algorithms) would increase the VRC PRISM Sharpe by 0.29. The VRC execution team reviews prime-brokerage commission schedules annually with this figure in mind. **Second**, the vol spread $\sigma_3 - \sigma_1$ is the highest-elasticity natural market parameter: wider regime

vol separation makes VRC’s regime-conditional edge more valuable. Markets with identifiable, wide-spread volatility regimes—large-cap US equity indices, major FX pairs—are therefore structurally better suited to the VRC PRISM strategy than low-spread markets. **Third**, particle count elasticity is near zero beyond $M = 10,000$, confirming that VRC’s production setting of $M = 20,000$ provides ample precision with no meaningful benefit from further increases.

14. The VRC PRISM Trading Protocol

This section documents the VRC derivatives desk’s operating procedures as implemented through PRISM. It is intended to serve as the authoritative reference for the VRC investment committee, compliance team, and any VRC analyst who manages a live PRISM position. All thresholds and parameters referenced below are VRC’s current production settings; they are reviewed quarterly by the VRC investment committee and updated in the PRISM parameter database.

14.1 VRC Coverage Universe

The VRC PRISM coverage universe consists of liquid options satisfying the VRC liquidity screen (Section 12) and is reviewed monthly by the VRC derivatives team. The current universe is:

Instrument Class	VRC Eligibility Criteria	VRC Universe Size
Index options (US)	SPX, NDX, RUT liquid monthly series	3 instruments
ETF options	SPY, QQQ, IWM	3 instruments
Single stocks (US)	ADV > \$2B, IV rank > 30, pass liquidity screen	30–60 names
International indices	DAX, FTSE 100, Nikkei 225 options	3 instruments

Single-stock names in the VRC universe are not fixed; the VRC derivatives team adds names when they pass all liquidity criteria and removes them when their IV rank falls below 20 for three consecutive months. Index options receive priority allocation from the VRC Kelly framework due to their superior liquidity and lower impact costs.

14.2 VRC Entry Conditions: The Four-Gate Rule

The VRC investment committee requires that **all four** of the following gates pass before the VRC derivatives desk may initiate any long gamma position. PRISM checks all four gates automatically at market open and flags eligible names to the

VRC analyst on duty. No VRC long gamma position may be opened without PRISM confirmation that all four gates are green.

- (1) **VRC VRP Signal Gate:** $z_t^{\text{VRP}} > z^* = 1.0$ (VRC regime-adjusted VRP in the top 16% of VRC historical distribution, as computed by the VRC HAR-RV-Regime model).
- (2) **VRC Arbitrage Condition Gate:** The PRISM structural arbitrage condition (31) holds for the current VRC posterior $\hat{\pi}_t^{\text{VRC}}$ —i.e., the gamma-weighted sum of regime variance differences is positive.
- (3) **VRC Sharpe Minimum Gate:** The pre-trade transaction-cost-adjusted Sharpe ratio (45), computed using VRC’s actual cost parameters, is ≥ 1.2 . This gate ensures VRC is not paying too much in execution costs relative to the expected gamma income.
- (4) **VRC Regime Certainty Gate:** VRC particle filter entropy $H(\hat{\pi}_t^{\text{VRC}}) \leq H_{\max} = 0.8$ (VRC filter is sufficiently confident in the current regime). This gate prevents VRC from entering a long gamma position during regime transition ambiguity when the entropy penalty would immediately reduce the position to a sub-economic size.

14.3 VRC Position Construction

Once all four VRC entry gates are green, the VRC derivatives analyst constructs the position using PRISM’s output:

- (1) **Option selection:** VRC buys the nearest monthly expiry ATM straddle (or ATM call for directional overlays approved by the VRC investment committee) with time to expiry $\tau \in [15, 45]$ trading days. Options with $\tau < 15$ days are excluded due to gamma-theta instability in the final week; options with $\tau > 45$ days are excluded due to lower gamma-per-dollar.
- (2) **VRC notional sizing:** The notional is set to $f_{\text{VRC}}^{\text{adj}} \times \text{VRC derivatives capital allocation}$, where $f_{\text{VRC}}^{\text{adj}}$ is from (48). The VRC investment committee sets the total derivatives capital envelope separately from the AEGIS equity book.
- (3) **VRC initial hedge:** The delta hedge is executed via E-mini futures at ratio (26) using the VRC VWAP schedule (59) if the required size exceeds 0.5% of ADV; otherwise executed at market via VRC’s prime broker.
- (4) **VRC trade record:** PRISM logs the inception implied vol $\hat{\sigma}_0$, regime posterior $\hat{\pi}_0^{\text{VRC}}$, notional, full greeks, and the pre-trade Sharpe estimate. This record is the baseline against which the VRC thesis integrity monitor tracks subsequent

P&L versus theoretical expectation.

14.4 VRC Daily Management Checklist

The VRC derivatives analyst on duty follows this PRISM-driven checklist every trading day for each active position.

VRC market open (9:30 AM ET):

- Run VRC particle filter update with overnight return and opening options chain snapshot (SPX and single-stock chains).
- Receive PRISM broadcast of updated $\hat{\pi}_t^{\text{VRC}}$, H_t^{VRC} , $\hat{\eta}_t^*$ via (38), and $f_{\text{VRC}}^{\text{adj}}$ via (48).
- Confirm all four VRC entry gates still pass for existing positions (if any gate fails, follow exit procedure below).

VRC intraday (continuous):

- PRISM monitors delta mismatch δ_t in real time.
- VRC execution desk rebalances futures hedge when $|\delta_t| \geq \hat{\eta}_t^*$ (PRISM alert sent to execution terminal).
- VRC particle filter weight update triggered automatically on any $\geq 10\%$ intraday move in the ATM IV of any VRC-held position.

VRC market close (4:00 PM ET):

- Run full VRC particle filter end-of-day update with official closing return and closing options chain.
- PRISM checks all five VRC exit conditions (Section 14.5) and flags any triggered conditions to the VRC investment committee.
- Flag futures roll requirement if $\tau \leq 5$ trading days on any held contract; schedule VRC two-day roll execution.

14.5 VRC Roll Protocol

VRC rolls all options positions when time to expiry falls below 5 trading days. The VRC two-day roll protocol minimises market impact and avoids the theta cliff of the final week:

Day 1 (VRC Roll Day 1): VRC execution desk closes 50% of the expiring position and simultaneously opens 50% of the new front-month position. Both legs are executed via VRC VWAP schedule if notional exceeds 1% ADV.

Day 2 (VRC Roll Day 2): VRC closes the remaining 50% of the expiring position

and completes the new position.

The VRC roll is never executed during the 30 minutes before or after a major macro release (CPI, FOMC, NFP). Roll timing is flagged by PRISM’s economic calendar module three days in advance. Roll cost is booked separately in the VRC P&L attribution system as “VRC roll slippage” and is tracked against the theoretical prediction from (30).

14.6 VRC Exit Conditions: The Five-Trigger Framework

The VRC derivatives analyst closes the position immediately (or within the next opening hour for large positions requiring VWAP) upon any of the following five triggers being activated by PRISM:

- (1) **VRC VRP Reversal Trigger:** VRC VRP z-score $z_t^{\text{VRP}} < -0.5$ for 3 consecutive VRC market closes. This signals that implied vol has fallen below VRC’s regime-adjusted realized vol estimate; the long gamma position is now a structural drain rather than a structural earner.
- (2) **VRC Crisis Regime Trigger:** VRC particle filter crisis probability $\hat{\pi}_{3,t}^{\text{VRC}} > 0.6$ *and* the position has produced negative realised P&L during the current regime transition. VRC does not exit a profitable crisis-regime long gamma position automatically—crisis vol spikes generate the largest single-day gamma income—but exits a loss-making one immediately.
- (3) **VRC Thesis Integrity Trigger:** Cumulative P&L falls more than $2.5\sigma_{\text{P\&L}}\sqrt{t}$ below PRISM’s theoretical mean (42). This is the VRC model misspecification alarm: if the realised P&L is more than 2.5 standard deviations below theoretical expectation, the VRC investment committee reviews whether the model parameters need re-estimation or the position premise has changed.
- (4) **VRC Expiry Trigger:** At $\tau = 0$; position is closed at expiry if not rolled per the VRC roll protocol above.
- (5) **VRC Hard Risk Limit Trigger:** Any hard limit in the VRC risk control table (below) is breached.

14.7 VRC Risk Controls: Hard and Soft Limits

The following risk limits are set by the VRC investment committee and enforced automatically by PRISM’s risk monitor. Hard limits trigger automatic position reduction or exit without analyst discretion; soft limits trigger a VRC alert requiring the analyst on duty to acknowledge and document their response within 30 minutes.

VRC Risk Factor	Metric	VRC Hard Limit	VRC Soft Limit
Vega concentration	Net vega/AUM	3% (force reduce)	2% (alert)
VRC regime entropy	$H(\hat{\pi}_t^{\text{VRC}})$	> 1.0 (exit all)	> 0.8 (reduce 50%)
Basis risk	$ b_t /S_t$	$> 4\sigma_b/\sqrt{2\kappa}$ (exit)	$> 2\sigma_b/\sqrt{2\kappa}$ (alert)
VRC tail risk	99% daily CVaR	0.75% notional	0.50% notional
Options liquidity	ATM bid-ask (vol)	$> 100\text{bp}$ (exit)	$> 75\text{bp}$ (reduce)
VRC leverage	Delta exposure/AUM	$2.0\times$ (halt)	$1.5\times$ (alert)
Portfolio vega corr	Cross-position vega corr	> 0.7 (halt new trades)	> 0.5 (review)

All VRC risk limit breaches are reported to the VRC investment committee within one hour and documented in the VRC risk management log. Repeated soft limit breaches on the same name (3 within 10 trading days) trigger a VRC investment committee review of that name’s continued eligibility for the PRISM coverage universe.

15. VRC PRISM Extensions and Roadmap

The following extensions represent VRC’s active research agenda for PRISM. Each is at a different stage of development, from theoretical framework (Merton-MRSGBM) to prototype testing (Heston-MRSGBM, options flow) to concept stage (cross-asset, RL). The VRC investment committee reviews the PRISM development roadmap quarterly alongside the AEGIS development roadmap.

15.1 VRC PRISM 2.0: Jump-Diffusion Regime Switching (Merton-MRSGBM)

The VRC derivatives desk has observed that the current MRSGBM underestimates the probability of large single-day moves around earnings announcements and macro events—a gap that the Merton-MRSGBM extension directly addresses. The extended VRC model adds a regime-dependent jump component to (6):

$$\frac{dS_t}{S_t} = (\mu_{X_t} - \bar{\lambda}_J m_J) dt + \sigma_{X_t} dW_t + (e^{J_t} - 1) dN_t^J, \quad (60)$$

where N_t^J is a Poisson process with VRC regime-dependent intensity $\lambda_J^{(i)}$ (calibrated by VRC to single-stock earnings histories), $J_t \sim \mathcal{N}(\mu_J, \sigma_J^2)$ is the log-jump size, and $m_J = e^{\mu_J + \sigma_J^2/2} - 1$ is the mean jump size. The VRC gamma scalping P&L identity (15) acquires an additional jump compensation term proportional to $\lambda_J^{(i)} \mathbb{E}[(V(Se^J) - V(S) - JSV_S)^2]$, which VRC’s PRISM 2.0 prototype computes analytically for log-

normal jumps. This extension is particularly important for VRC’s single-stock PRISM positions around earnings, where jump intensity $\lambda_j^{(i)}$ spikes by an order of magnitude in the 48 hours around announcements.

15.2 VRC PRISM 2.1: Stochastic Volatility Hybrid (Heston-MRSGBM)

VRC trades VIX futures as a portfolio-level hedge alongside its single-name PRISM positions. To price and hedge VIX derivatives consistently within the PRISM framework, the VRC research team is developing a Heston-MRSGBM hybrid that integrates stochastic volatility within each regime:

$$dS_t = \mu_{X_t} S_t dt + \sqrt{v_t} S_t dW_t^S, \quad (61)$$

$$dv_t = \kappa_v(\bar{v}_{X_t} - v_t) dt + \xi \sqrt{v_t} dW_t^v, \quad (62)$$

with $dW_t^S dW_t^v = \rho_v dt$. This VRC “regime-switching Heston” model captures both the within-regime stochasticity of volatility (Heston dynamics, which generates the VRC-observed vol-of-vol) and between-regime jumps in the long-run variance level ($\bar{v}_{X_t}$, the regime-specific mean-reversion target). The VRC research team expects this model to improve SPX ATM vol fitting by approximately 12% on an out-of-sample RMSE basis relative to the current VRC MRSGBM, based on preliminary VRC prototype testing.

15.3 VRC PRISM 2.2: Options Market Flow Intelligence

The VRC derivatives desk already monitors order flow imbalance from VRC’s Bloomberg TOWR feed in real time. The PRISM 2.2 roadmap item integrates this data formally into the VRC particle filter as a fourth likelihood component. The VRC signed options volume imbalance signal is:

$$\Phi_t^{\text{VRC}} = \frac{(\text{Call vol} \times \Delta) - (\text{Put vol} \times \Delta)}{\text{Total signed delta volume}} \quad (63)$$

which VRC monitors at the portfolio level daily. A large positive Φ_t^{VRC} (heavy call buying) tends to suppress realised vol in the next session as market makers’ long-gamma hedging dampens intraday moves [Hull, 2014], while a large negative Φ_t^{VRC} (heavy put buying) tends to amplify moves. VRC’s preliminary analysis on SPX options flow data from 2018–2024 shows that $|\Phi_t^{\text{VRC}}| > 1.5\sigma_\Phi$ predicts a same-direction realised vol change with 61% accuracy in the following session, making it a useful auxiliary signal for the VRC intraday bandwidth update.

15.4 VRC PRISM 2.3: Cross-Asset Regime Integration

The VRC macro research team, which covers global credit markets for the AEGIS macroeconomic regime module [Verma, 2026a], has documented that CDX investment-grade and high-yield spreads lead VRC equity volatility regime transitions by 3–5 trading days on average [Granger and Poon, 2003]. VRC’s PRISM 2.3 roadmap item will incorporate a bivariate HMM fitted jointly to VRC equity and VRC credit data:

$$\Lambda^{\text{VRC,cross}} = \Lambda^{\text{equity}} \otimes \Lambda^{\text{credit}}, \quad (64)$$

producing a richer, more forward-looking VRC regime signal. This cross-asset layer will be shared between AEGIS and PRISM, allowing VRC to maintain a single regime posterior across both its equity and derivatives books.

15.5 VRC PRISM 3.0: Reinforcement Learning Bandwidth Optimisation

The VRC HJB bandwidth formula (37) is derived under parametric assumptions that are known to be approximations. VRC’s longest-horizon roadmap item is a deep reinforcement learning layer that learns the optimal VRC bandwidth function directly from simulated P&L, without imposing the parametric structure of the HJB solution:

$$\eta_{\text{VRC}}^*(s) = \arg \max_{\eta} Q^{\text{VRC}}(s, \eta; \theta), \quad (65)$$

where $s = (S, \delta, b, \hat{\pi}^{\text{VRC}}, t)$ is the full VRC state vector and $Q^{\text{VRC}}(s, \eta; \theta)$ is the VRC action-value function parameterised by neural network weights θ , trained on VRC-calibrated MRSGBM simulations. The VRC investment committee will require that PRISM 3.0 passes a six-month paper-trading validation against the HJB formula before any live deployment on the VRC derivatives book.

16. VRC Limitations and Risk Disclosures

The VRC investment committee requires that every VRC whitepaper include a candid and complete discussion of limitations. PRISM is a powerful and carefully constructed framework, but the VRC team does not regard any model as a substitute for judgment. The limitations below are maintained as a live document; the VRC investment committee reviews them whenever a model parameter update or a significant drawdown event triggers a PRISM thesis integrity alert.

16.1 VRC Model Risk

PRISM is built on the MRSGBM, which is a purposeful simplification of market dynamics. The VRC model assumes that regime transitions occur instantaneously—in reality, markets can spend days transitioning smoothly between volatility environments. The model also assumes within-regime log-normality, whereas VRC’s own return data shows excess kurtosis even within single regimes. A third, subtler risk is the feedback between VRC’s own hedging activity and the underlying price: if VRC’s delta hedge is large enough to affect the E-mini market, the quadratic variation process $\langle L \rangle_t$ becomes endogenous and the P&L identity (15) must be modified. At VRC’s current AUM this feedback is negligible, but the VRC investment committee monitors it as the derivatives book grows. The VRC thesis integrity exit condition (Trigger 3 in Section 14.5) is specifically designed to detect model misspecification and alert VRC before losses accumulate.

16.2 VRC Liquidity Risk

The VRC optimal bandwidth formula (37) and VWAP execution model (59) both assume that VRC can execute any required hedge at the expected cost within the execution window. During market dislocations—which are also the times when VRC’s long gamma positions are most valuable—bid-ask spreads can widen abruptly and the E-mini market can become temporarily shallow. VRC’s hard liquidity limit (>100 bp ATM bid-ask triggers exit) is designed to prevent VRC from being trapped in an illiquid position, but it necessarily means VRC may exit some crisis-regime positions before the full gamma windfall is captured.

16.3 VRC Regime Detection Latency

The VRC particle filter detects regime shifts with a mean lag of 1.7 trading days (Table 4). During this lag, the VRC derivatives book may be sized for the wrong regime. The VRC crisis detection latency of 2.6 days is the most consequential, as it means VRC may carry a calm-regime-sized long gamma position through the initial, most violent phase of a vol spike. The VRC entropy penalty (48) partially compensates by automatically shrinking the VRC position during high-uncertainty transition periods, but it cannot fully eliminate the detection lag. The VRC investment committee has accepted this latency as an unavoidable cost of the probabilistic approach and has calibrated the VRC crisis hard exit trigger ($\hat{\pi}_{3,t}^{\text{VRC}} > 0.6$) to balance early exit against false alarms.

16.4 VRC Correlation Risk

When market-wide volatility spikes, every VRC PRISM position benefits simultaneously—and when vol compresses, every position bleeds theta simultaneously. This intra-portfolio correlation creates a fat-tailed VRC P&L distribution that is not fully captured by the per-position Sharpe ratio metric. The VRC risk control table enforces a portfolio vega correlation hard limit of 0.7, and the VRC investment committee reviews the portfolio-level vega correlation matrix weekly. VRC partially mitigates correlation risk by maintaining PRISM positions across multiple sectors and across the VRC international universe (DAX, FTSE, Nikkei), since international volatility regimes are imperfectly correlated with the US regime even during global stress events.

16.5 VRC Behavioural and Execution Discipline

Even a theoretically sound strategy can underperform due to behavioural overrides. The VRC investment committee has identified one specific pattern as the dominant cause of VRC derivatives underperformance in pre-PRISM manual gamma scalping: the tendency to *widen* the hedge bandwidth during drawdown periods (“giving the trade more room”), which is precisely the wrong response according to the VRC HJB analysis. The optimal VRC bandwidth widens only when $\sigma_i^2 > \hat{\sigma}_i^2$ (the edge is large), not when the P&L is temporarily negative. PRISM enforces mechanical bandwidth discipline by removing analyst discretion from the rebalancing decision entirely: the VRC execution desk rebalances when and only when PRISM broadcasts a rebalance signal. The VRC compliance framework requires that any deviation from a PRISM bandwidth signal be documented and justified to the VRC investment committee within one business day.

17. Conclusion

This paper has introduced PRISM, the Probabilistic Regime-Integrated Scalping Model, the proprietary framework that governs VRC’s equity derivatives arbitrage book. PRISM is not a theoretical exercise; it is the operational backbone of VRC’s derivatives desk, and every equation in this paper corresponds to a live computation that influences a real VRC trading decision.

The key contributions that VRC has developed and validated are as follows.

First, the VRC regime-switching P&L identity (Theorem 4.1), which decomposes

the VRC derivatives desk’s gamma scalping P&L into a gamma-weighted average of regime-conditional variance differences. This decomposition reveals that the VRC effective breakeven implied vol $\hat{\sigma}^*$ is biased toward high-vol regimes, meaning that any VRC analyst using a flat historical realised vol as their breakeven will systematically underestimate the strategy’s expected profitability—and therefore undersize their position.

Second, the VRC three-asset no-arbitrage characterisation (Theorems 6.2–6.3), which provides the precise conditions under which the VRC options-futures-spot system admits a riskless profit and derives the annualised arbitrage P&L rate α^{arb} in closed form. This rate is the pre-trade P&L target that VRC books against every PRISM position.

Third, the VRC basis-enhanced arbitrage result (Corollary 6.4), which proves that the VRC futures-hedging approach provides a strictly positive additional P&L stream through basis mean-reversion carry. VRC estimates this enhancement at 15–25 bps per year of annualised notional in the current carry environment.

Fourth, the VRC regime-adjusted optimal bandwidth formula (Theorem 7.1), which gives VRC’s derivatives desk a mathematically precise answer to one of the most practically important questions in gamma scalping: when should you rebalance? The VRC answer: when the delta mismatch exceeds $(3c\sigma_i/\Gamma_i S)^{1/3} \cdot [1 + (\sigma_i^2 - \hat{\sigma}_i^2)/3\sigma_i^2]$, and not before.

Fifth, the VRC Fokker-Planck P&L distribution analysis (Section 8), establishing that VRC’s gamma scalping P&L is right-skewed and providing the Cornish-Fisher correction for accurate VRC VaR reporting to the investment committee.

Sixth, the VRC multi-signal particle filter (Algorithm 1), which achieves 91.3% regime identification accuracy with a mean lag of 1.7 trading days by combining VRC return-based, VRC realised-vol-based, and VRC options-surface-based likelihoods. The options surface channel is VRC’s proprietary addition and accounts for 4.4% of the overall accuracy improvement.

Seventh, the VRC Kelly-entropy position sizing framework (equation (48)), which gives every VRC PRISM position a single, model-derived, automatically updated size that reflects both the current expected edge *and* the current regime uncertainty. VRC considers this the most operationally valuable output of the entire framework: it removes discretionary position sizing from the derivatives desk and replaces it with a principled, reproducible, committee-approved formula.

The VRC Monte Carlo evidence (Table 3) confirms that the VRC PRISM adaptive framework delivers a Sharpe ratio of 2.42, versus 1.34 for naïve spot hedging—an 81%

improvement—with a probability of profit of 79.6% per path. The VRC investment committee considers the combination of these numbers, together with the right-skew of the P&L distribution and the theoretical grounding of every parameter, to constitute a robust and defensible basis for deploying VRC capital in systematic gamma scalping.

The VRC derivatives book will continue to be managed exclusively through PRISM. Every new PRISM position is pre-approved by a four-gate PRISM check. Every active PRISM position is monitored by a five-trigger PRISM exit framework. Every VRC risk limit is enforced automatically by PRISM’s risk monitor. That is what it means to have a *system*: not a set of guidelines to consult when convenient, but a set of rules that govern every decision.

At VRC, gamma is not a static edge. It is a regime-conditional, continuously updated probability that volatility will exceed its implied level. PRISM is the architecture that makes that probability actionable—and VRC is the team that deploys it.

A. VRC Mathematical Derivations

A.1 Proof of Proposition 3.3 (VRC Regime-Mixing IV Correction)

Expand $V_i = V^{\text{BS}}(S, t; \bar{\sigma}) + \epsilon_i U_i(S, t)$ where $\epsilon_i = \sigma_i^2 - \bar{\sigma}^2$ and U_i satisfies:

$$\mathcal{L}^{\text{BS}} U_i = -\frac{1}{2} S^2 V_{SS}^{\text{BS}} + \frac{1}{\epsilon_i} \sum_{j \neq i} \lambda_{ij} U_j, \quad (66)$$

with $\mathcal{L}^{\text{BS}}$ the Black-Scholes operator. The ATM Black-Scholes vega satisfies $V_\sigma^{\text{BS}}|_{\text{ATM}} = S\sqrt{\tau/2\pi}$. Matching the ϵ^2 correction and using $\epsilon_1 = -\pi_2/\pi_1 \cdot \epsilon_2$ gives (13). $\square$

A.2 Proof of Theorem 7.1 (Optimal Bandwidth)

In the no-rebalancing region, the HJB (35) in the δ direction alone reduces (at leading order) to:

$$\frac{1}{2} \sigma_i^2 S^2 \Gamma_i^2 J_{\delta\delta} = -\frac{1}{2} \sigma_i^2 S^2 \Gamma_i (\sigma_i^2 - \hat{\sigma}_i^2). \quad (67)$$

This gives $J_{\delta\delta} = -(\sigma_i^2 - \hat{\sigma}_i^2)/\Gamma_i$, hence $J \approx J_0 - \frac{(\sigma_i^2 - \hat{\sigma}_i^2)}{2\Gamma_i} \delta^2$. Smooth pasting $J_\delta|_{\eta^*} = -cS$ gives the naïve estimate $\eta_0^* = cS\Gamma_i/(\sigma_i^2 - \hat{\sigma}_i^2)$. Including the leading Brownian variance correction for δ (which scales as $\sigma_i S \Gamma_i$) recovers the cubic-root formula via a balance-of-powers argument. $\square$

A.3 VRC Stationary Moments of the Basis Process

Under Assumption 5.2, the n th absolute moment of b_∞ is:

$$\mathbb{E}[|b_\infty|^n] = \frac{2^{n/2}\Gamma((n+1)/2)}{\sqrt{\pi}} \left(\frac{\sigma_b^2}{2\kappa}\right)^{n/2}, \quad (68)$$

in particular $\mathbb{E}[|b_\infty|] = \sigma_b\sqrt{2/(\pi\kappa)}$. Substituting into (28) gives the expected VRC annual basis income:

$$\text{BI}^{\text{VRC}} = (r - q)\bar{h}^F \sigma_b\sqrt{2/(\pi\kappa)}, \quad (69)$$

where $\bar{h}^F$ is the time-averaged VRC futures hedge ratio. At VRC's current production parameters ($r - q = 3.45\%$, $\sigma_b = 0.08\%$, $\kappa = 12$), this evaluates to approximately 1.1 bps per year per unit of futures hedge ratio—a small but consistent additive contribution to VRC's gamma scalping P&L.

B. VRC PRISM Prompt Library (LLM Integration)

PRISM integrates a VRC LLM layer (currently Claude Sonnet, consistent with VRC's AEGIS LLM infrastructure [Verma, 2026a]) for monitoring unstructured market commentary and news that may provide early warning signals of regime transitions. These prompts are part of the VRC PRISM production system and are reviewed quarterly by the VRC derivatives research team. They are presented here as part of the VRC institutional record.

B.1 VRC Volatility Regime Narrative Assessment Prompt

You are an expert quantitative analyst at Verma Research Capital (VRC) specializing in equity volatility regimes. The following are the past 5 days of macro news headlines and options market commentary for [TICKER]. VRC uses a three-regime MRSGBM model calibrated to S&P 500 data: Regime 1 (VRC Calm, RV ~ 10%), Regime 2 (VRC Normal, RV ~ 22%), Regime 3 (VRC Crisis, RV ~ 45%). Your task is to:

- (1) Assess whether the current information is consistent with VRC Regime 1, 2, or 3.
- (2) Identify any early warning signals of a VRC regime transition.
- (3) Rate your confidence from 1-10 for each regime assignment.
- (4) Flag any information that conflicts with VRC's current particle filter posterior estimate [CURRENT_VRC_POSTERIOR].

Return a structured JSON with fields: vrc_regime_posterior,

vrc_transition_signal, vrc_confidence, vrc_conflicts, and supporting_evidence (verbatim source passages only).

B.2 VRC Earnings Volatility Risk Assessment Prompt

You are analyzing [TICKER] for the VRC PRISM derivatives book, ahead of earnings release in [N] days. VRC's current PRISM position on this name is: [POSITION_SUMMARY]. Using the provided analyst consensus estimates, live options chain data (attached), and the company's last 8 earnings call summaries (attached), assess:

- (1) The probability that realized post-earnings vol exceeds implied vol (VRC VRP signal direction).
- (2) The most likely direction of the earnings surprise and its expected magnitude in vol terms.
- (3) Whether the current options pricing reflects the historical VRC-observed earnings move distribution for this name.
- (4) Whether VRC should reduce, maintain, or increase the existing PRISM position ahead of the earnings event.

Return a JSON with: vrc_vol_surprise_prob, vrc_direction_bias, vrc_pricing_assessment, vrc_position_recommendation, and supporting_evidence.

C. VRC PRISM Live Coverage Universe Snapshot

Table 6 presents a representative snapshot of the VRC PRISM coverage universe as assessed by PRISM on a typical market day. All entries reflect the VRC particle filter regime posterior, VRC HAR-RV-Regime VRP z-score, and VRC pre-trade Sharpe at the time of the snapshot. **Active** positions are on the VRC derivatives book; **Watch** names have passed the VRC Sharpe and arbitrage condition gates but not yet the VRP z-score gate; **Avoid** names have failed the VRP gate in crisis regime.

Table 6: VRC PRISM Coverage Universe (Representative VRC Snapshot)

Instrument	VRC Regime	VRC VRP Z	Arb Rate (bps/yr)	VRC SR	VRC Status
SPX (Index)	VRC Calm	+1.82	112	2.71	VRC Active
QQQ (ETF)	VRC Calm	+1.51	98	2.44	VRC Active
NVDA	VRC Normal	+0.94	67	1.89	VRC Watch
AAPL	VRC Calm	+1.23	81	2.12	VRC Active
TSLA	VRC Normal	+0.41	39	1.31	VRC Neutral
AMZN	VRC Calm	+1.67	93	2.38	VRC Active
GS	VRC Normal	+0.78	55	1.67	VRC Watch
DAX (Intl)	VRC Normal	+0.62	47	1.54	VRC Watch
META	VRC Calm	+1.44	87	2.21	VRC Active
JPM	VRC Crisis	−0.91	−34	−0.87	VRC Avoid

D. VRC PRISM Glossary

PRISM Probabilistic Regime-Integrated Scalping Model. VRC’s proprietary derivatives arbitrage framework, the derivatives complement to AEGIS within the VRC research infrastructure.

MRSGBM Markov Regime-Switching Geometric Brownian Motion. The VRC spot price model defined in Section 3.

VRC Calm / Normal / Crisis VRC’s three production regimes with realised vol parameters $\sigma_1 = 10.2\%$, $\sigma_2 = 22.1\%$, $\sigma_3 = 44.8\%$, calibrated to S&P 500 2000–2024 data.

Gamma (Γ) Second derivative of option price with respect to spot. The central driver of VRC gamma scalping P&L.

VRC VRP Variance Risk Premium: VRC’s regime-adjusted measure of how much the market is overpaying for volatility relative to VRC’s HAR-RV-Regime forecast.

VRC VRP Z-Score (z_t^{VRP}) VRC’s primary entry signal: the normalised distance of the current VRP from its regime-conditional historical mean. VRC enters when $z > 1.0$.

Delta Mismatch (δ_t) The current difference between the VRC option delta and the VRC futures hedge position. PRISM triggers rebalancing when $|\delta_t| \geq \hat{\eta}_t^*$.

VRC Optimal Bandwidth ($\hat{\eta}_t^*$) The PRISM-computed Whalley-Wilmott-VRC threshold for the delta mismatch. Broadcast to the VRC derivatives desk at market open and on intraday triggers.

Basis (b_t) Deviation of the VRC-held futures price from theoretical cost-of-carry. Mean-reverts with speed κ ; generates VRC basis income via (28).

VRC Particle Filter VRC's multi-signal SIR particle filter (Algorithm 1) combining returns, Garman-Klass realised vol, and VRC options surface likelihood.

VRC Four-Gate Rule VRC's entry prerequisite: all four gates (VRP, arbitrage condition, Sharpe minimum, regime certainty) must be green before any VRC long gamma position is initiated.

VRC Five-Trigger Framework VRC's exit rules: any of five PRISM-monitored triggers closes the VRC position.

VRC Kelly-Entropy Size ($f_{\text{VRC}}^{\text{adj}}$) VRC's automatically computed position fraction, from (48). Smaller when VRC regime entropy is high; full half-Kelly when regime is certain.

HJB Equation Hamilton-Jacobi-Bellman equation; the VRC stochastic control foundation for the optimal bandwidth derivation.

VRC Thesis Integrity Monitor PRISM's continuous comparison of VRC realised P&L against theoretical expectation (42). Triggers exit and committee review when $> 2.5\sigma$ below expectation.

HAR-RV-Regime VRC's Heterogeneous Autoregressive model of Realised Variance extended with regime-conditional coefficients. Used to compute the VRC VRP z-score.

AEGIS VRC's Autonomous Equity Grading and Intelligence System [Verma, 2026a]. The equity valuation complement to PRISM within VRC's integrated research infrastructure.

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